Typing Odia Phonetically

A reference sheet: what to type in English to get ଓଡ଼ିଆ

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Pitfall. This sheet is a **v0.1 draft scheme**, derived from ITRANS conventions and published Odia orthography references. It has **not been reviewed by a native Odia speaker**. Treat every mapping as a proposal, not a fact. Where it is wrong, fix it in `core/rules/symbols.toml` — that file is the single source of truth, and this sheet should be regenerated to match.

1 How it works

Odia is an *abugida*, not an alphabet. That one fact explains most of the rules below.

Key idea. Every consonant already carries a built-in vowel. କ on its own is `ka`, not `k`. To attach a different vowel you add a **matra** (a vowel sign). To attach *no* vowel — so the consonant joins the next one — the script uses a **halant**, which produces a conjunct.

There are two typing modes.

Natural mode (default)
: Type it the way it sounds. Case is ignored. Several spellings reach the same word, and the engine offers a ranked candidate list — you pick. This is what you will use day to day.

Strict mode (`--strict`)
: Full ITRANS. Case matters. Exactly one deterministic answer, no candidates. Use it when you need a specific character and do not want to argue with the ranking.

In the tables below, the **Type** column is natural mode and **Strict** is the ITRANS form. Where they are identical, Strict is left blank.

2 Vowels

Each vowel has two shapes: an **independent** letter, used at the start of a word or after another vowel, and a **matra**, used after a consonant. You type the same thing for both — the engine picks the right shape from context.

Independent	On କ	Type	Strict	Example
ଅ	କ	`a`		ଅଛ `achha`
ଆ	କା	`aa`	`A`	ଆମ `aama`
ଇ	କି	`i`		ଇଏ `ie`
ଈ	କୀ	`ii`, `ee`	`I`	ନଈ `naii`
ଉ	କୁ	`u`		ଉଠ `utha`
ଊ	କୂ	`uu`, `oo`	`U`	ମୂଳ `muula`
ଋ	କୃ	`ru`, `ri`	`R^i`	ଋଷି `rushi`
ଏ	େକ	`e`		ଏକ `eka`
ଐ	ୈକ	`ai`, `oi`		ଐକ୍ୟ `aikya`
ଓ	ୋକ	`o`		ଓଠ `otha`
ଔ	ୌକ	`au`, `ou`		ଔଷଧ `aushadha`

Pitfall. **Double the vowel to lengthen it.** This is the single most common reason a word comes out wrong. `a` is ଅ but `aa` is ଆ, so ଭାଗ୍ୟ is `bhaagya`, not `bhagya`. Likewise `ii`/`ee` for ଈ and `uu`/`oo` for ଊ. Frequent words are recognised either way, but for anything else the doubling matters.

Note. The rare vowels ୠ `RR`, ଌ `Lu` and ୡ `LL` exist in the scheme but almost never appear in modern Odia. They are available in strict mode only.

3 Consonants

The five *vargas* (articulation groups). Read each row left to right: unaspirated, aspirated, voiced, voiced-aspirated, nasal.

	Position in the mouth				
Group	voiceless		voiced		nasal
Velar	କ `ka`	ଖ `kha`	ଗ `ga`	ଘ `gha`	ଙ `nga`
Palatal	ଚ `cha`	ଛ `chha`	ଜ `ja`	ଝ `jha`	ଞ `nya`
Retroflex	ଟ `Ta`	ଠ `Tha`	ଡ `Da`	ଢ `Dha`	ଣ `Na`
Dental	ତ `ta`	ଥ `tha`	ଦ `da`	ଧ `dha`	ନ `na`
Labial	ପ `pa`	ଫ `pha`	ବ `ba`	ଭ `bha`	ମ `ma`

Pitfall. **Retroflex vs dental** is the one distinction English does not have, and the main reason you will see multiple candidates. Retroflex (ଟ ଠ ଡ ଢ ଣ) is made with the tongue curled back; dental (ତ ଥ ଦ ଧ ନ) with the tongue on the teeth. In strict mode a **capital** letter means retroflex: `Ta` is ଟ, `ta` is ତ. In natural mode case is ignored and both are offered — the dictionary decides which is more likely.

Everything else

Letter	Type	Notes	Letter	Type	Notes
ଯ	`ya`	sounds like *ja*	ଶ	`sha`	palatal
ର	`ra`		ଷ	`Sha`	retroflex
ଲ	`la`		ସ	`sa`	dental
ଳ	`La`	retroflex *l*	ହ	`ha`	
ୱ	`ba`, `va`		ଓ	`wa`	ସ୍ୱର `swara`

The three special letters

These carry a *nukta* (a dot below) or are otherwise distinct. They matter more in Odia than in most Indic scripts — the first one appears in the name of the language.

Letter	Type	Sound	Example
ଡ଼	`rr`, `d`	retroflex flap	ଓଡ଼ିଶା `odishaa`, ବଡ଼ `bada`
ଢ଼	`rh`, `dh`	aspirated flap	ପଢ଼ା `paRhaa`
ୟ	`y` (not word-initial)	true *y* sound	ରାଜ୍ୟ `raajya`

Pitfall. ଯ and ୟ are different letters, and Odia uses them the opposite way round from Hindi. ଯ is pronounced like *ja*; ୟ is the one that actually sounds like *y*. So typing `y` gives ଯ at the *start* of a word (`yoga` → ଓଯୋଗ) and ୟ anywhere else (`raajya` → ରାଜ୍ୟ, `nyaaya` → ନ୍ୟାୟ). Both stay in the candidate list, so you can override either way.

4 Signs and marks

These are combining marks, so they are shown here attached to କ — on their own they would be almost invisible.

Mark	On କ	Name	Type	Example
ଂ	କଂ	anusvara	`n`, `m`, `M`	ଅଂଶ `ansha`
ଁ	କଁ	candrabindu	`n~`, `Mn`	ମୁଁ `mun`
ଃ	କଃ	visarga	`H`	ଦୁଃଖ `duHkha`
◌୍	କ୍	halant	automatic	see below

Anusvara nasalises *before a following consonant*; candrabindu nasalises the vowel itself; visarga is a breathy release; halant strips the built-in vowel so a conjunct can form.

5 Conjuncts

When two consonants meet with no vowel between them, they fuse into a single conjunct glyph. **You do not type the halant** — just type the consonants back to back and the engine forms the conjunct.

Type	Get	Type	Get	Type	Get
`kta`	କ୍ତ	`pra`	ପ୍ର	`nta`	ନ୍ତ
`ksha`	କ୍ଷ	`shra`	ଶ୍ର	`Nda`	ଣ୍ଡ
`jny`	ଜ୍ଞ	`swa`	ସ୍ୱ	`mbha`	ମ୍ଭ
`tra`	ତ୍ର	`nka`	ଙ୍କ	`ddha`	ଦ୍ଧ
`sta`	ସ୍ତ	`ShTa`	ଷ୍ଟ	`lla`	ଲ୍ଲ
`gya`	ଗ୍ୟ	`nya`	ନ୍ୟ	`dya`	ଦ୍ୟ

Key idea. **A typed `n` is five different letters**, depending on what follows it, because a nasal takes the place of articulation of the next consonant: `n`+`k/g` → ଙ (ଙ୍କ, ଗଙ୍ଗ); `n`+`ch/j` → ଞ (ପଞ୍ଚମ); `n`+`T/D` → ଣ (ମୁଣ୍ଡ); `n`+`t/d` → ନ (ଅନନ୍ତ); `n`+`p/b` → ମ (ସମ୍ବାଦ). The anusvara ଂ is offered alongside all of them (ସଂସ୍କାର). So the honorific is simply `nka`: `baabunka` → ବାବୁଙ୍କ.

Pitfall. `ny` and `gy` are read as *two* letters, not as ଞ and ଜ୍ଞ, because in real words that is nearly always what they are --- `dhanyabaada` → ଧନ୍ୟବାଦ, `bhaagya` → ଭାଗ୍ୟ. For the single letters type `~n` → ଞ and `jny` → ଜ୍ଞ.

Note. To force a bare consonant with a visible halant and *no* conjunct — which you occasionally want at the end of a word — append an underscore: `kamal_` gives କମଲ୍ rather than କମଲ.

6 Numerals

Type	0	1	2	3	4	5	6	7	8	9
Get	୦	୧	୨	୩	୪	୫	୬	୭	୮	୯

Odia numerals are **off by default** — modern written Odia overwhelmingly uses Arabic digits. Turn them on with the `digits = "odia"` setting.

7 Worked examples

Type this	Get this	Meaning
`namaskaara`	ନମସ୍କାର	hello
`kemiti achha`	କେମିତି ଅଛ	how are you
`dhanyabaada`	ଧନ୍ୟବାଦ	thank you
`mun bhala achhi`	ମୁଁ ଭଲ ଅଛି	I am well
`odishaa`	ଓଡ଼ିଶା	Odisha
`odiaa`	ଓଡ଼ିଆ	Odia (the language)
`bhubaneswara`	ଭୁବନେଶ୍ୱର	Bhubaneswar
`paani`	ପାଣି	water
`tume`	ତୁମେ	you
`bhaagya`	ଭାଗ୍ୟ	fate, luck
`raajya`	ରାଜ୍ୟ	state
`nyaaya`	ନ୍ୟାୟ	justice

Key idea. You are not expected to spell these correctly on the first try. The engine offers a ranked list, you choose, and it remembers your choice. Accuracy on the words *you* personally use climbs quickly — that is the whole design.

8 Five things that trip people up

1. **Double the vowel to lengthen it.** By far the most common mistake. ଭାଗ୍ୟ is `bhaagya`; `bhagya` would be ଭଗ୍ୟ, which is not a word. Same for `ii`/`ee` and `uu`/`oo`.

2. **The built-in vowel is not silent.** ଭଲ is `bhala`, not `bhal`. It is also pronounced closer to *o* than *a* — which is why ଓଡ଼ିଶା is written "Odisha" in English, not "Adisha".

3. **ଡ଼ is not ଡ.** The flap ଡ଼ is extremely common in Odia (ବଡ଼, ପଢ଼ା, ଓଡ଼ିଶା). Internally it is always stored as ଡ plus a separate nukta character, never as a single precomposed character — they look identical on screen but do not match each other in a search.

4. **ଳ is a real letter,** not a typo for ଲ. Retroflex *l*, common in Odia: ମୂଳ, ଫଳ, କାଳ. Type `La`.

5. **Case only matters in strict mode.** In natural mode `Ta` and `ta` are the same input and you disambiguate by picking a candidate.

9 Fixing this sheet

Every mapping above corresponds to an entry in `core/rules/symbols.toml`. When a common word comes out wrong, adjust the weight there rather than working around it:

```
[[consonant]]
roman = "s"
odia  = [
    { text = "ସ", w = 1.0 },
    { text = "ଶ", w = 0.5 },
    { text = "ଷ", w = 0.4 },
]
```

Raising a weight promotes that character in the candidate ranking. After any change, run `npm run`

`eval` and confirm the accuracy delta against the committed baseline before keeping it — a fix for one word often costs you three others.